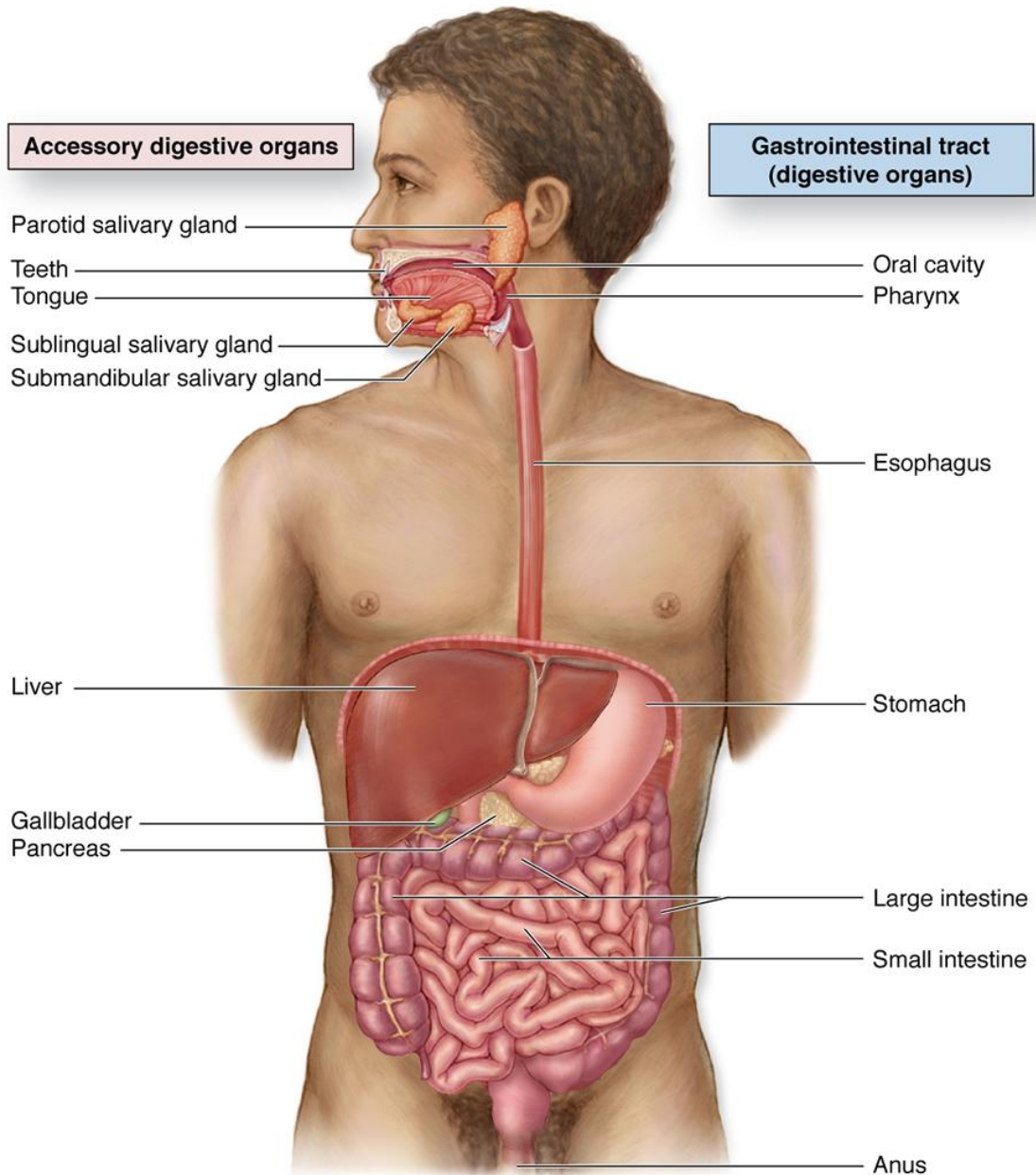


Digestive system



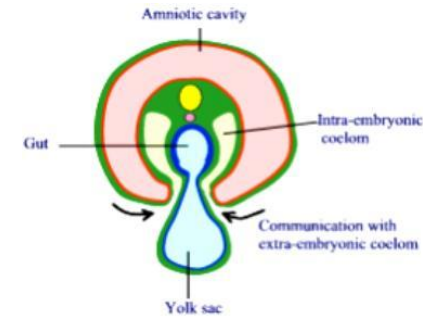
The **peritoneum** is the largest serous membrane of the human body.

Derives from lateral plates of mesoderm.

It has two layers:

1. parietal peritoneum
2. visceral peritoneum

They have a continuous mesothelial layer



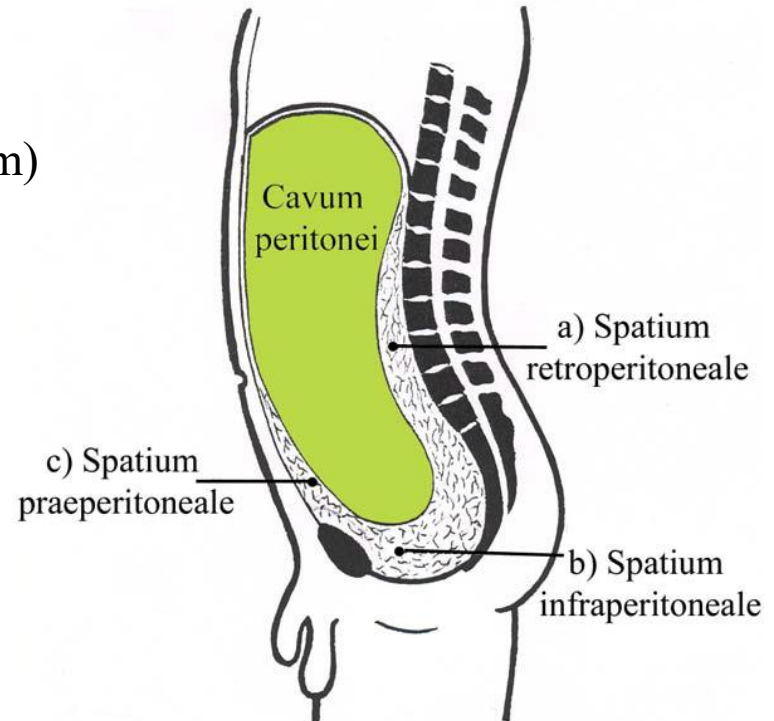
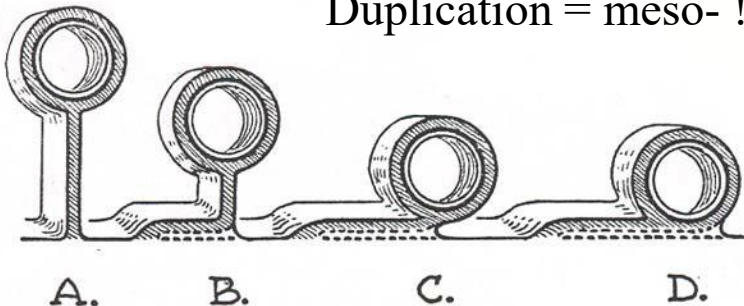
Abdominal and pelvic organs and structures –
according to their peritoneal relations - can be divided into 2 main groups:

Intraperitoneal - covered by peritoneum
and generally they have a duplication

Extraperitoneal (not surrounded by the peritoneum)
within this:

- Praeperitoneal
- Retroperitoneal (secunder and primary)
- Infracperitoneal structures.

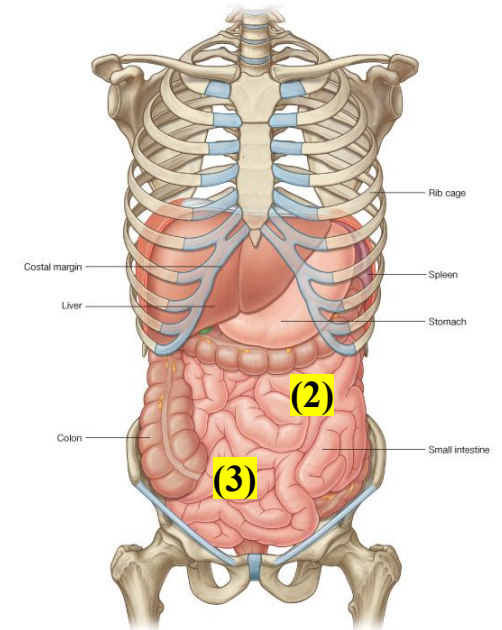
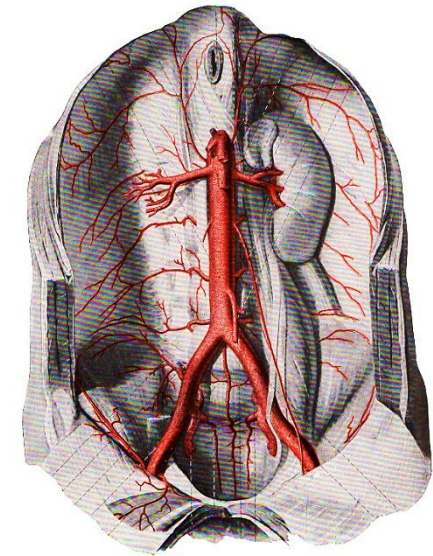
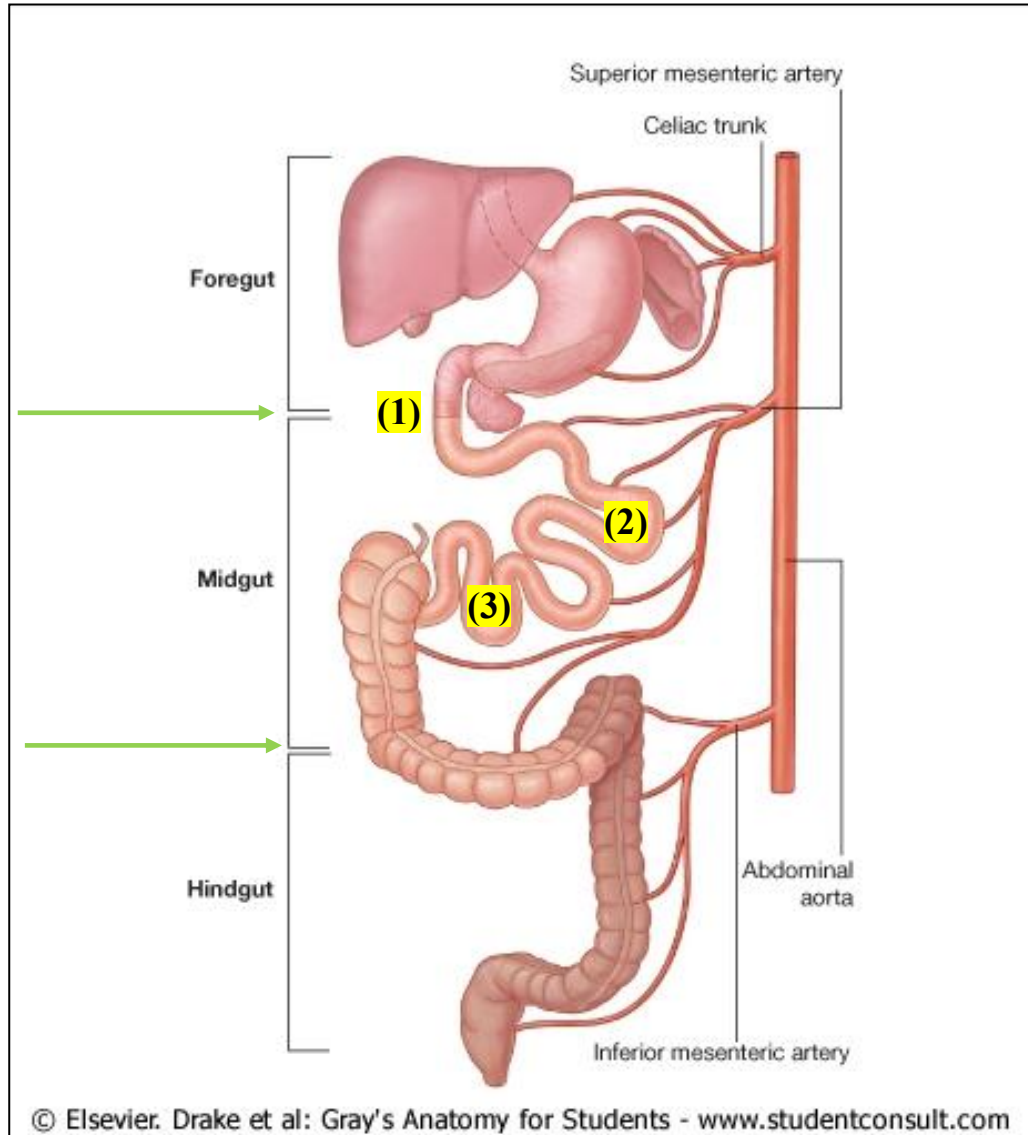
Duplication = meso- !!!!



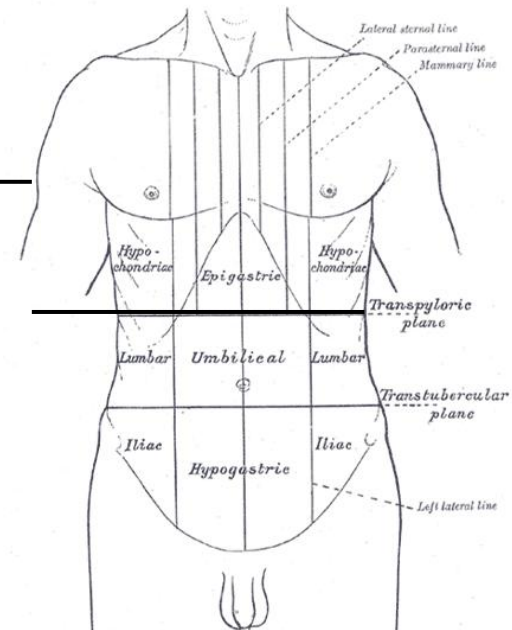
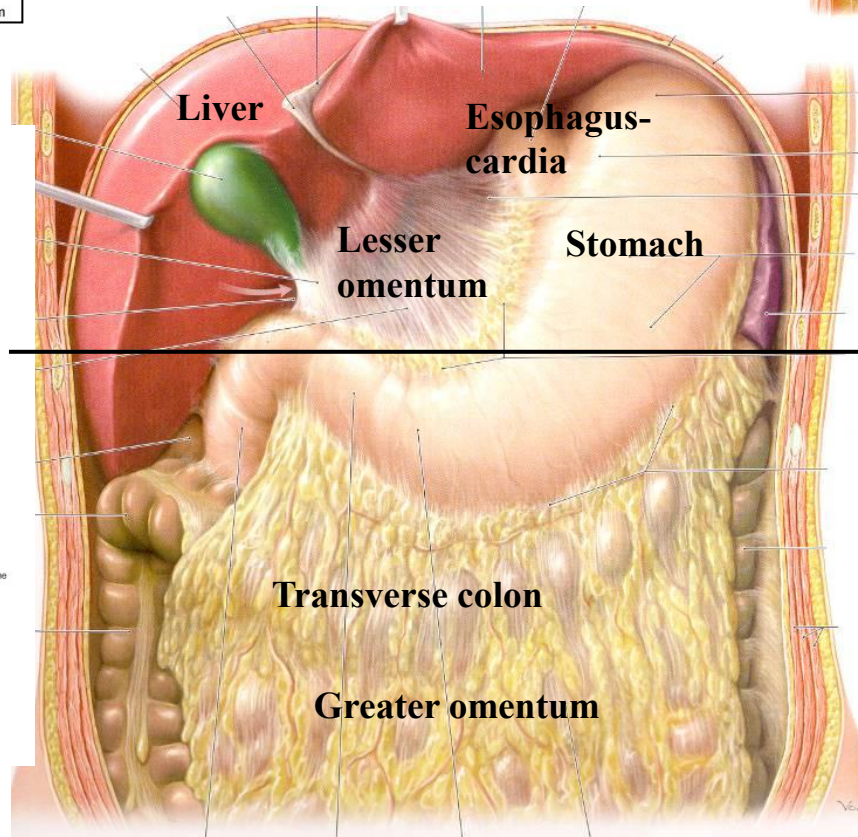
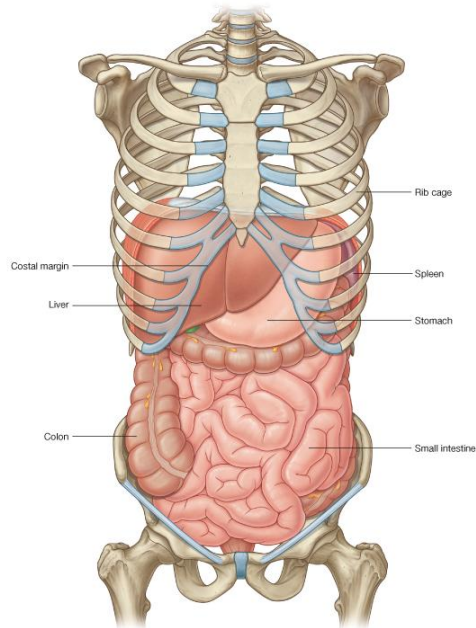
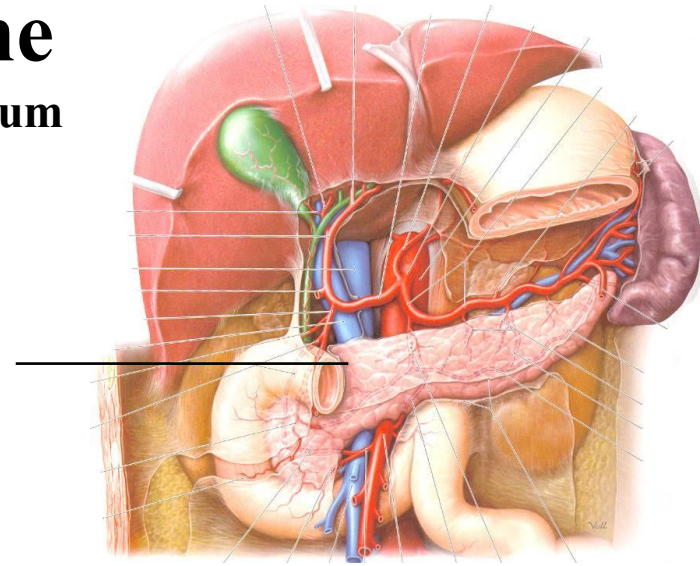
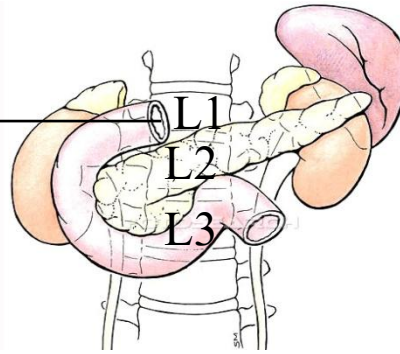
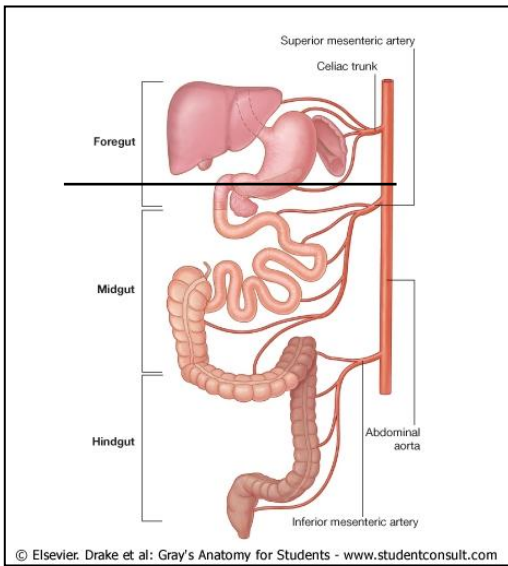
Abdominal part of the digestive system

developmental origin and blood supply (unpaired visceral branches of the abdominal aorta)

Small intestine: 1) duodenum, 2) jejunum, 3) ileum



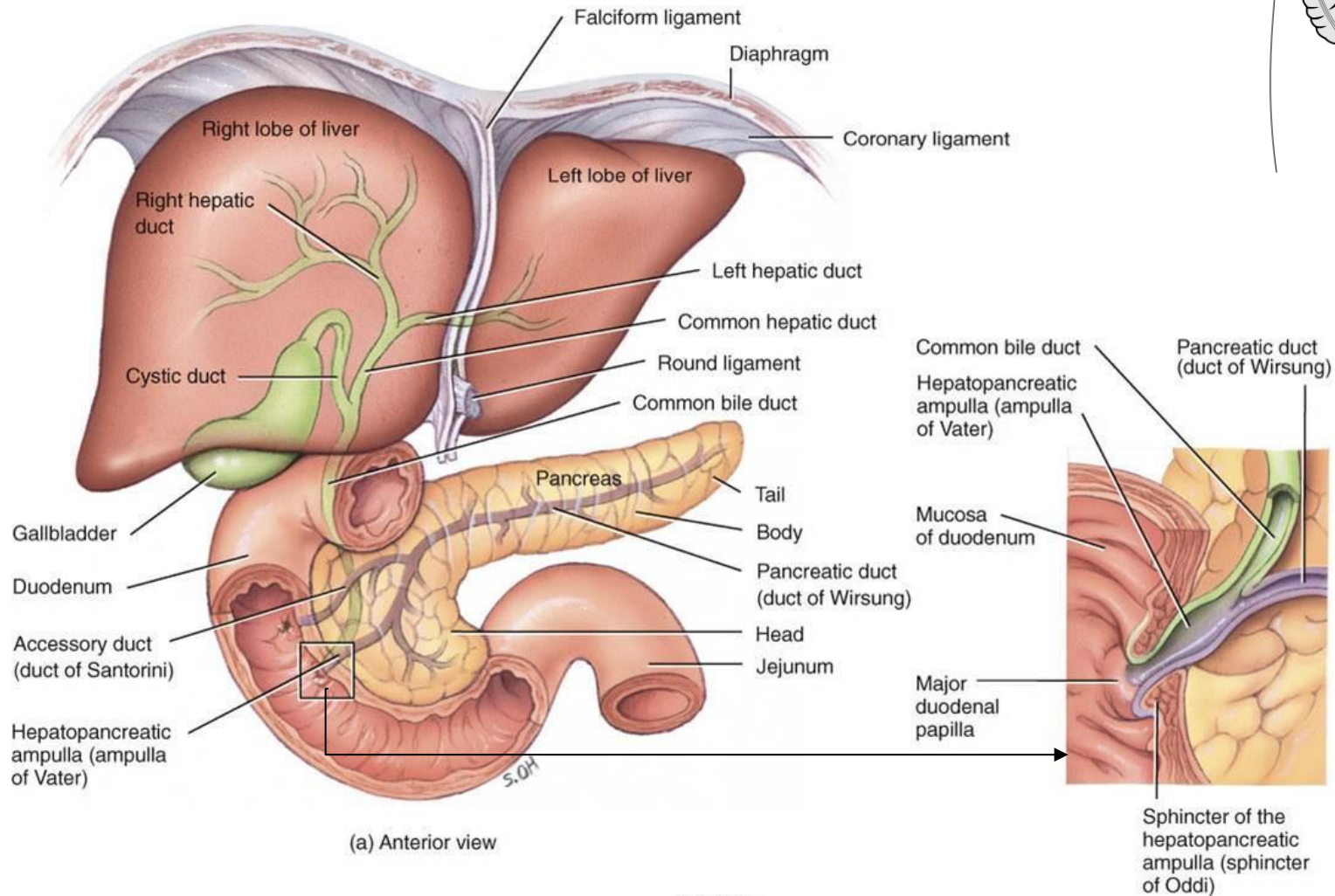
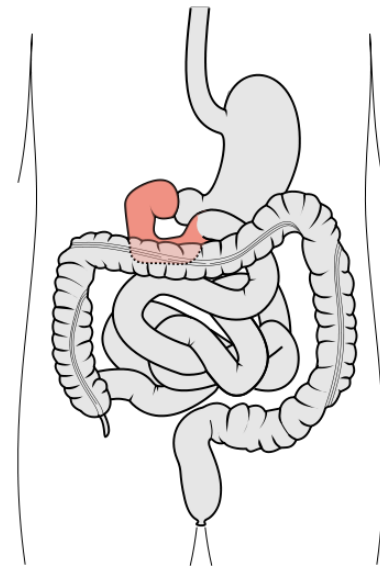
Transpyloric plane and the position of the duodenum



Duodenum

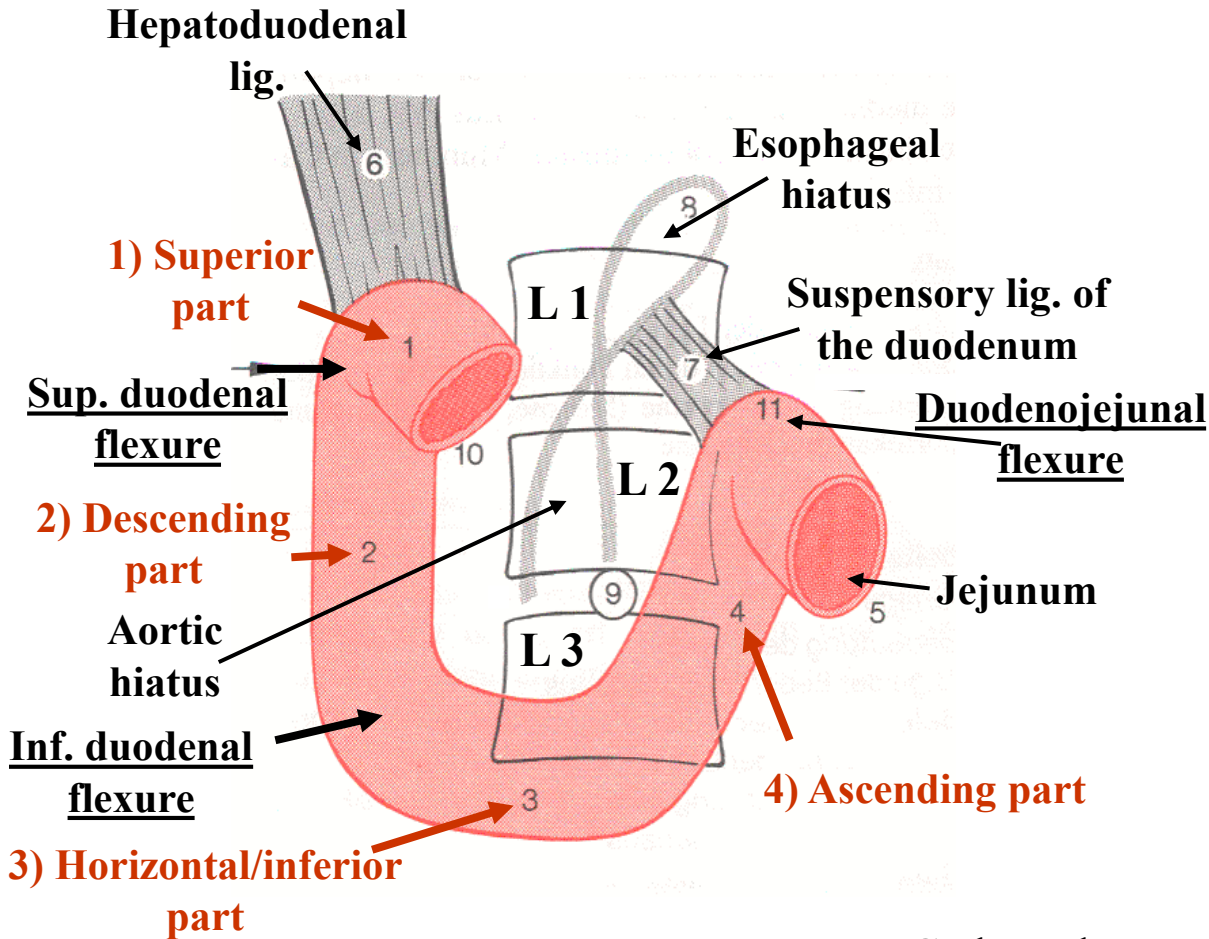
The duodenum is the proximal, U-shaped, and shortest (approx. 25–30 cm) segment of the small intestine, interconnecting the pylorus with the jejunum. *The name derived from the Latin *interstinum duodenum digitorum* ("12 fingers interstine").*

It acts as the primary site for mixing chyme with **bile** and **pancreatic enzymes** for nutrient digestion. Neutralizes stomach acid and initiates nutrient absorption.



(a) Anterior view

Parts of the duodenum and topography



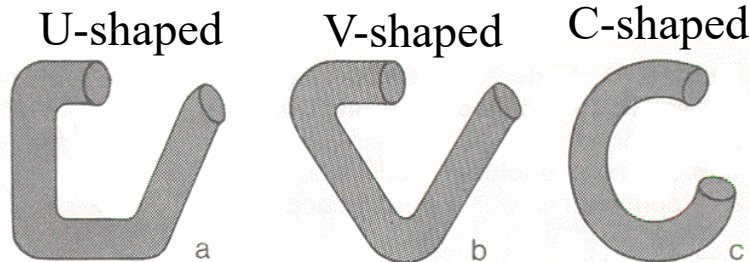
Total length: 25-30 cm;

1) Superior part: initial segment=bulbus (or ampulla); moveable; intraperitoneal (hepatoduodenal lig.);

2) Descending part: fixed; retroperitoneal;

3) Horizontal/inferior part: fixed; retroperitoneal;

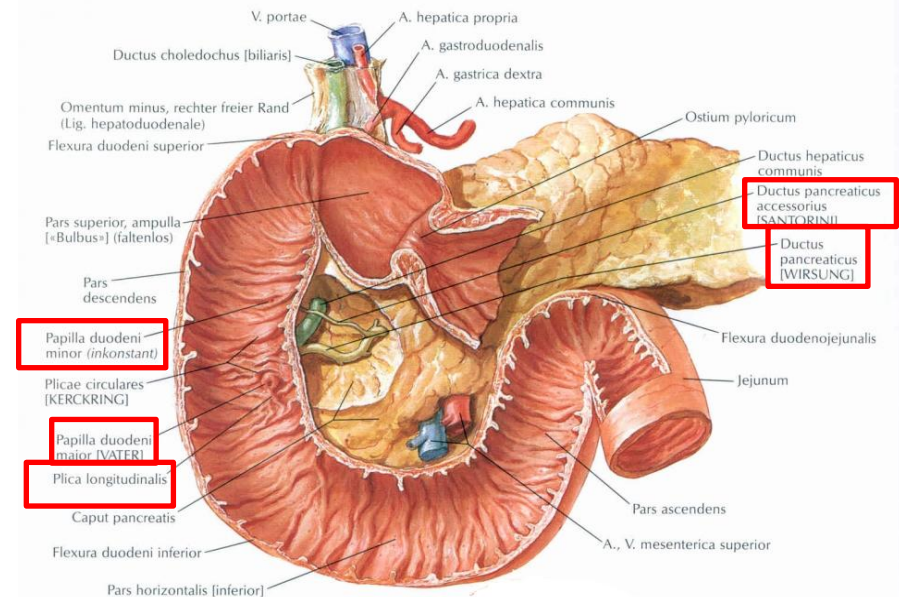
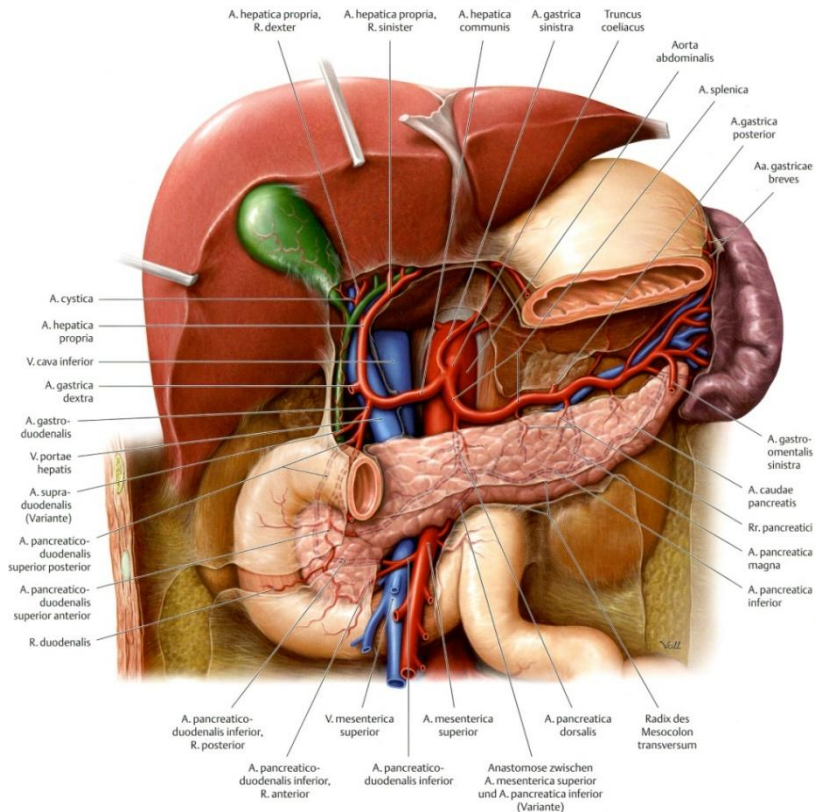
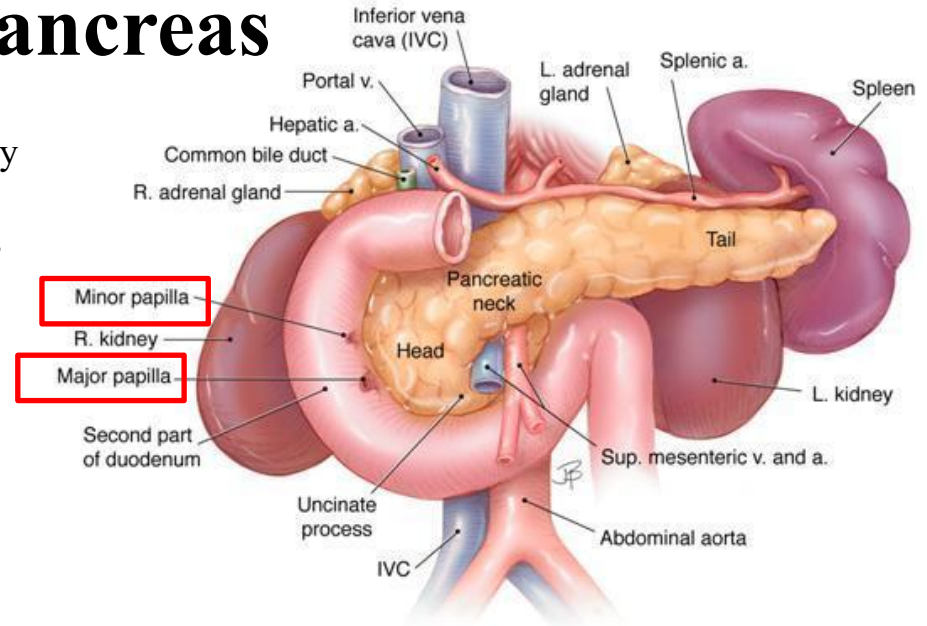
4) Ascending part: retro-, and intraperitoneal; The suspensory muscle and ligament suspends it to the origin of the sup. mesenteric artery.



Connections of the duodenum, the bile tract and the pancreas

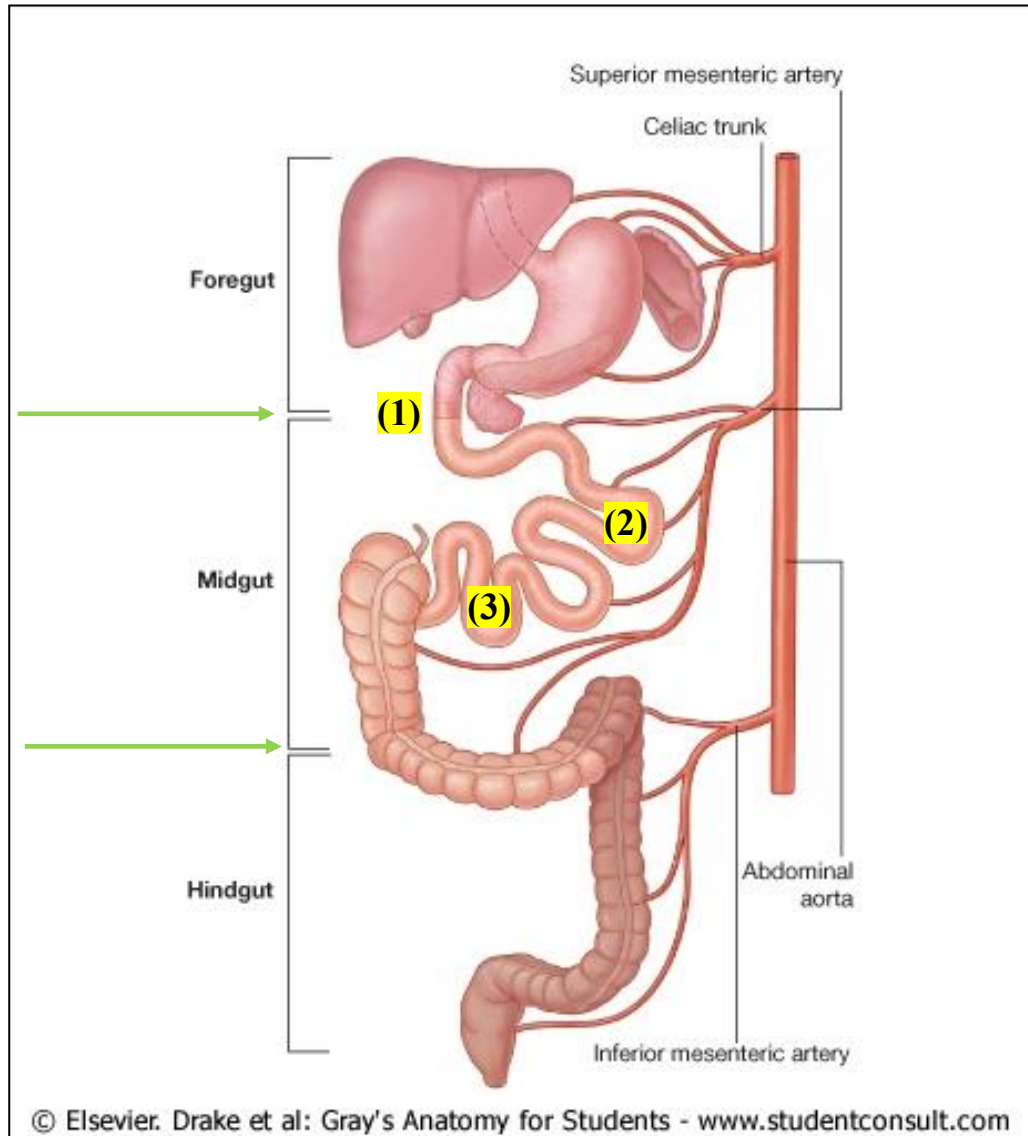
The **major duodenal papilla (papilla of Vater)** is a small, nipple-like structure in the duodenum that acts as the primary entrance for **bile and pancreatic juice**, surrounded by the sphincter of Oddi, it regulates the flow of digestive enzymes into the duodenum.

The **minor duodenal papilla** serves as the entrance for the **accessory pancreatic duct**.



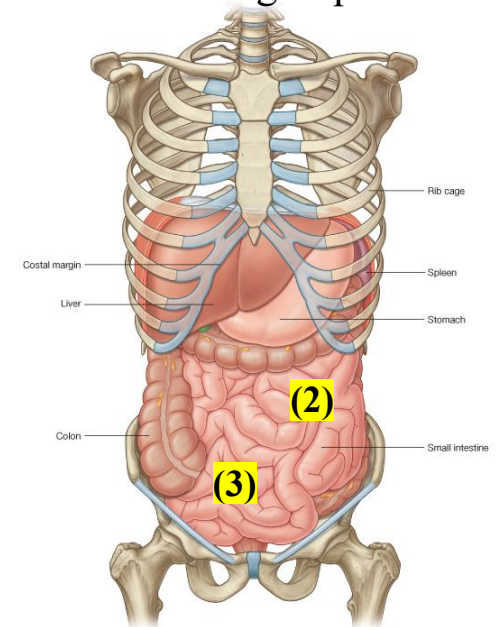
Small intestine – jejunum and ileum

Small intestine: 1) duodenum, 2) jejunum, 3) ileum



The (2) **jejunum** and (3) **ileum** are the two longest parts of the small intestine following the duodenum, measuring roughly **6–7 meters in total, form loops** and are responsible for nutrient absorption.

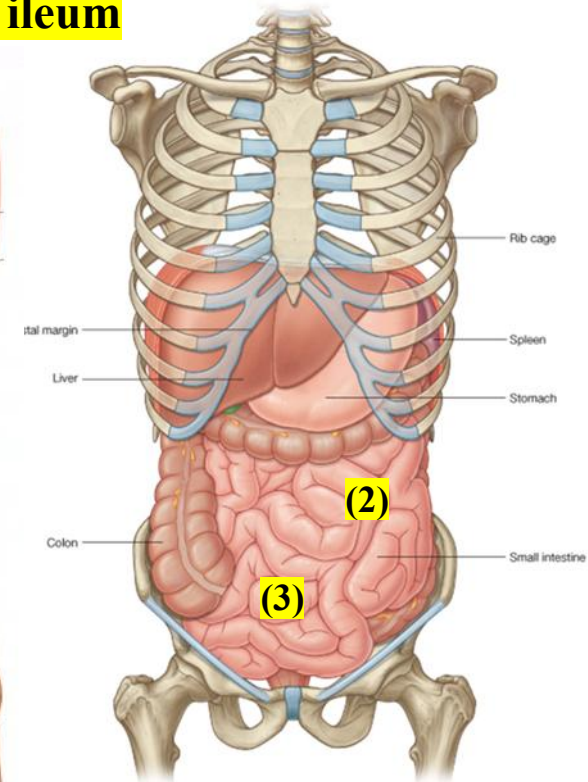
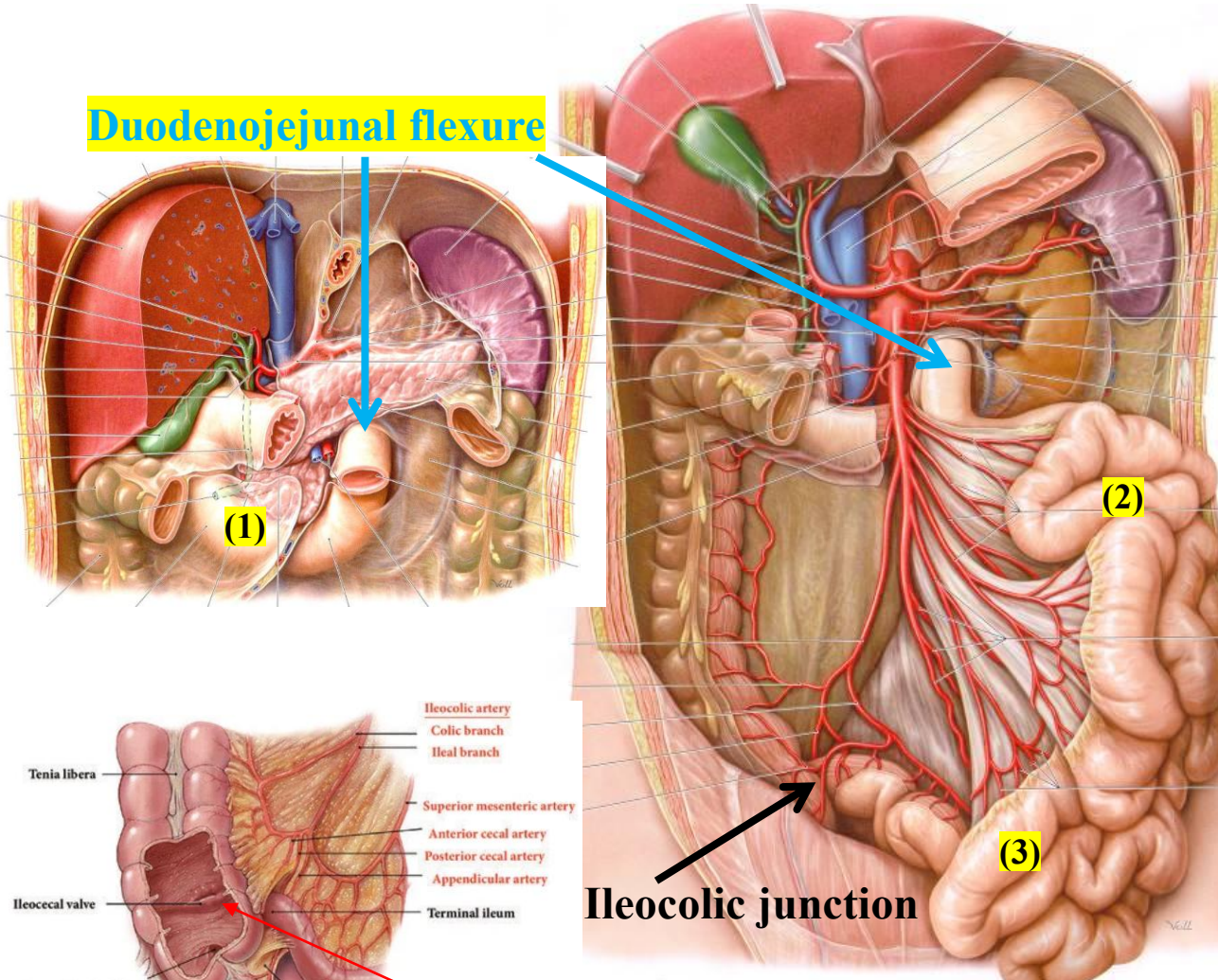
Location: The jejunum is situated in the upper left quadrant, while the ileum is in the lower right quadrant.



Duodenojejunal flexure and ileocolic junction

Small intestine: 1) duodenum, 2) jejunum, 3) ileum

Duodenojejunal flexure

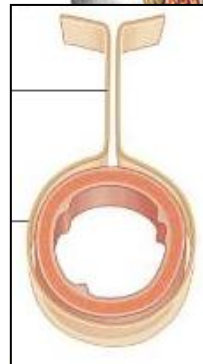
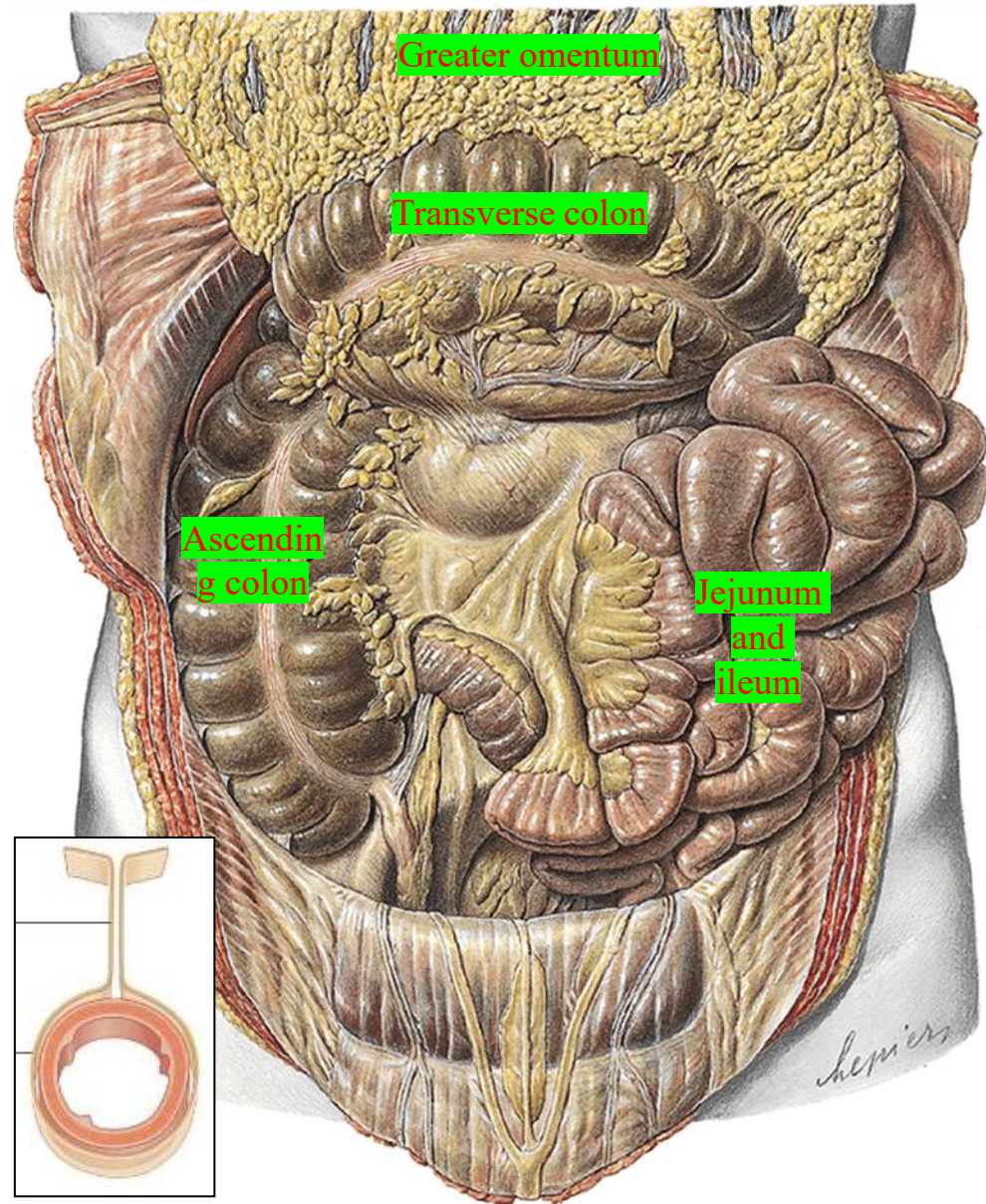
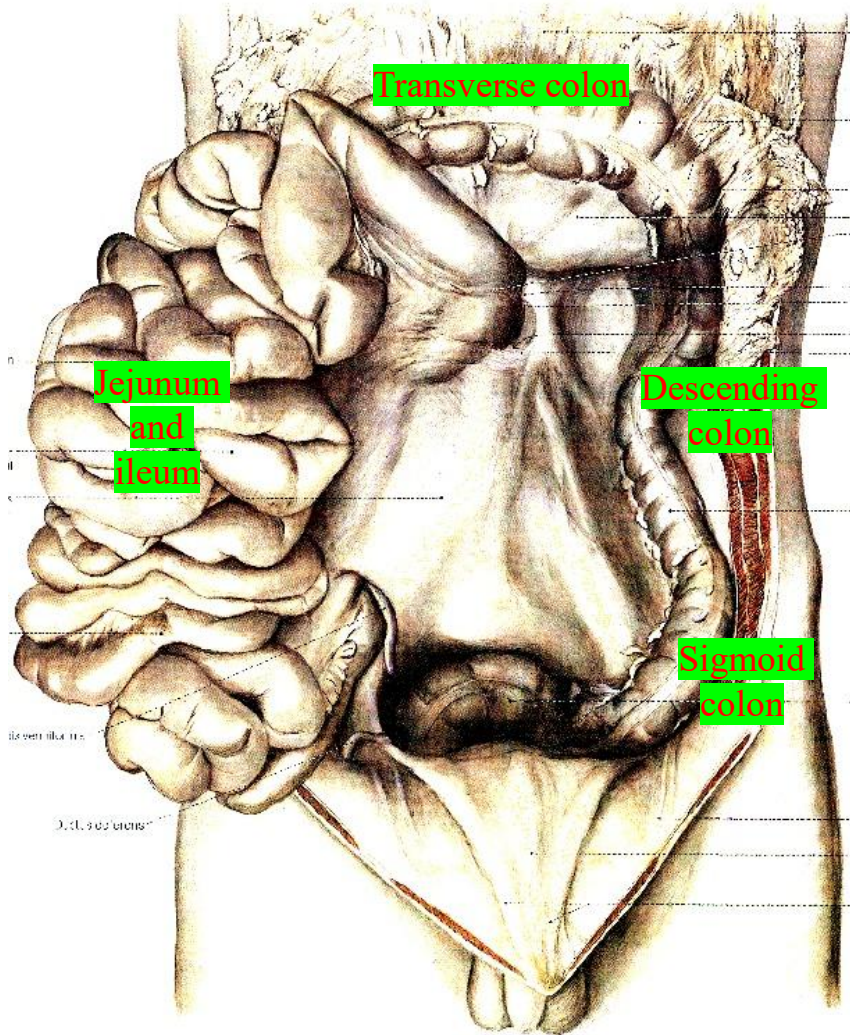


Ileocolic junction

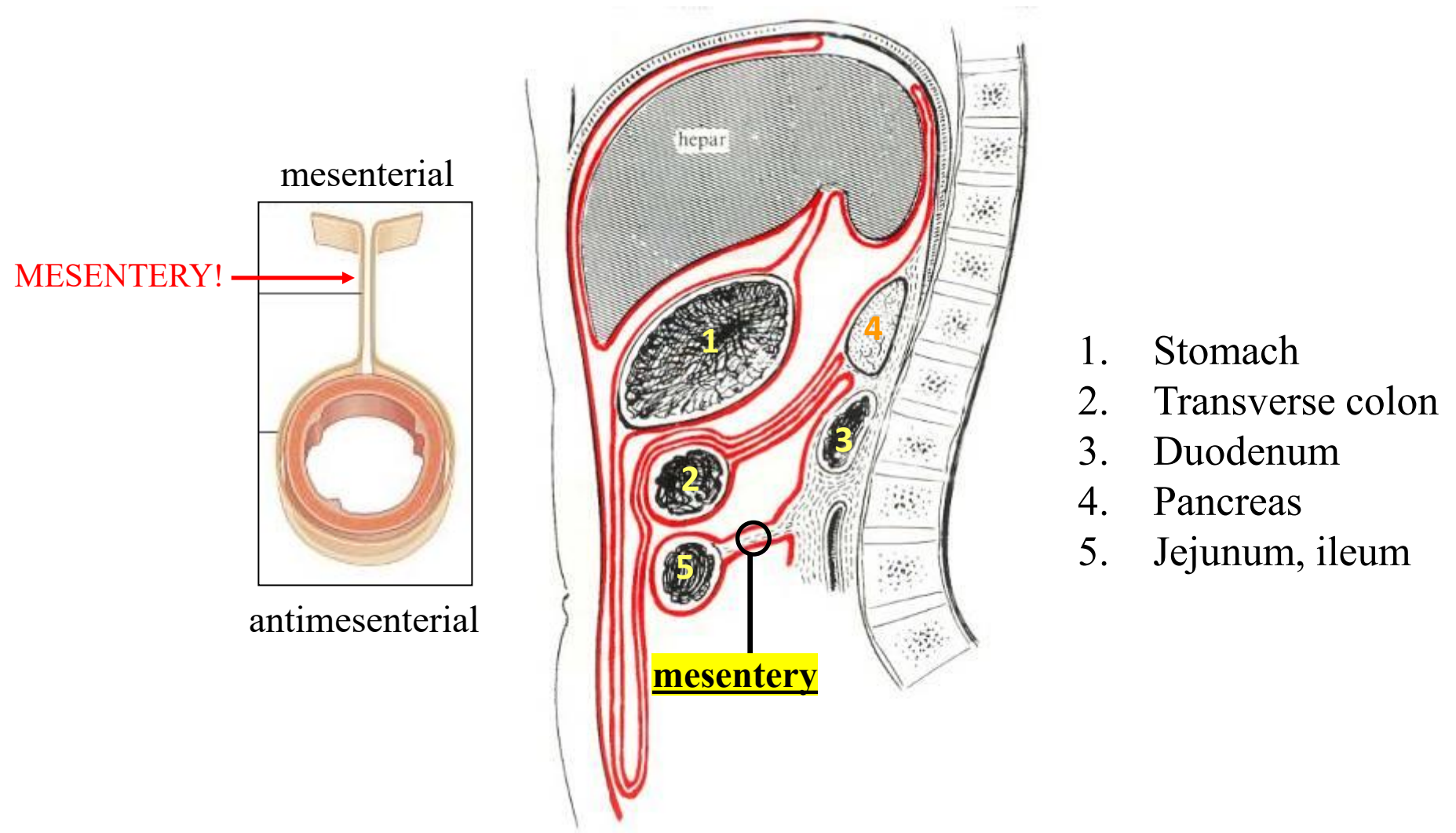
Bauchin's ileocecal valve

The **ileocecal valve** is a sphincter muscle situated at the junction of the small intestine (ileum) and large intestine (cecum/colon). It regulates the flow of contents from the ileum to the cecum and prevents reflux of colonic contents back into the small intestine.

Jejunum and ileum are intraperitoneal organs

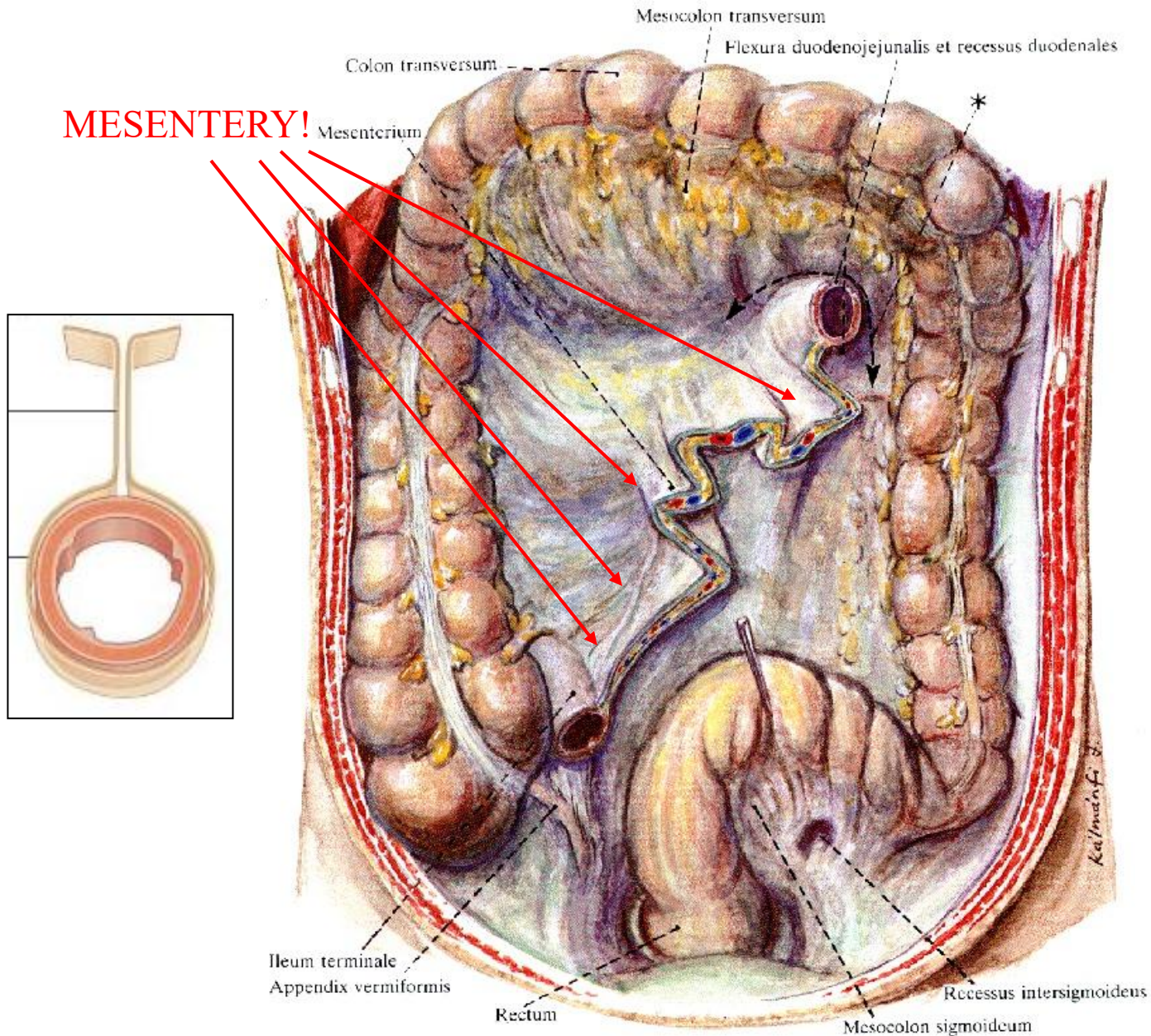


Jejunum and ileum are intraperitoneal organs



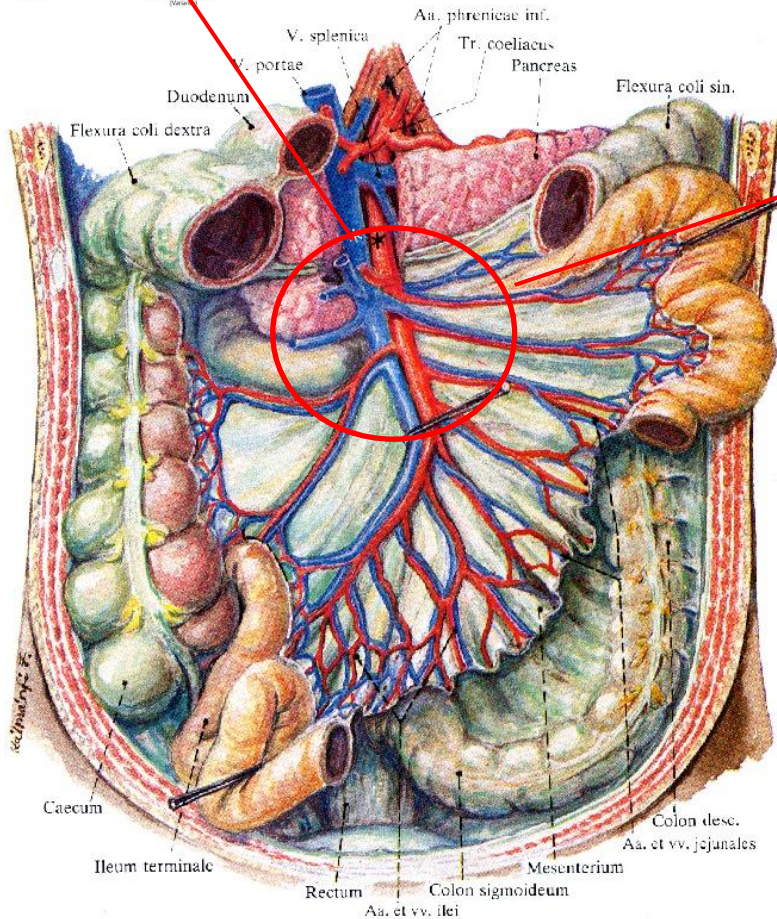
Midsagittal section of the abdominal cavity

Mesentery (root of the mesentery)



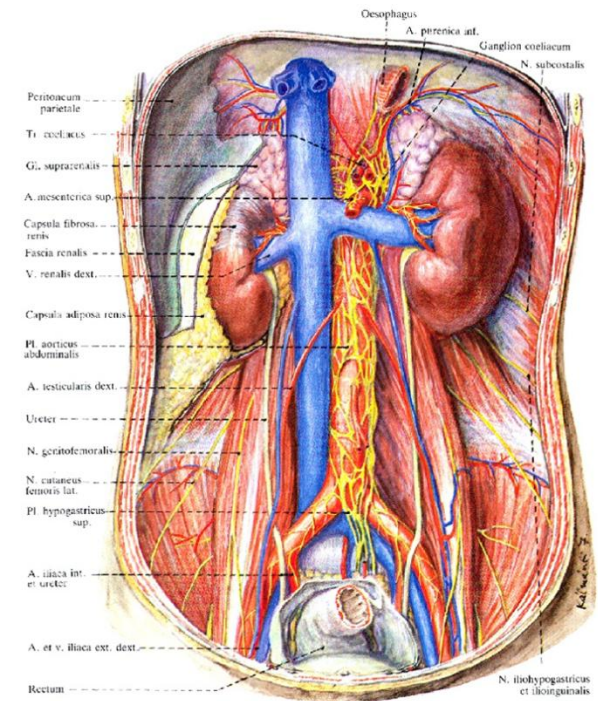
[illegible]

This anatomical diagram illustrates a cross-section of the human abdomen at the L3 vertebral level. The central feature is the vertebral body of L3, surrounded by the intervertebral discs and the spinal canal. The abdominal cavity is divided into several compartments by the peritoneum. Key structures labeled include the kidneys (Ren.), adrenal glands (M. suprarenalis), and the abdominal aorta (A. cava inf.). The diagram also shows the sympathetic trunk (Tr. sympathicus), the quadratus lumborum muscle (M. quadratus lumborum), and the descending colon (Colon desc.). The transverse colon (Colon transversum) is shown with its associated mesocolon. The superior mesenteric artery (A. mesenterica sup.) and its branches are visible, along with the celiac ganglion (G. coeliacum) and the celiac plexus (Plex. coeliacus). The inferior mesenteric artery (A. mesenterica inf.) and its branches are also shown. The diagram includes labels for the pancreas (Pars horizontalis duodeni), the omentum majus et cavitas peritonealis, and the mesenterium. The abdominal wall muscles, including the external oblique (M. rectus abdominis) and the internal oblique (M. obliquus int.), are also depicted. A red arrow points to the celiac ganglion, which is located anterior to the T12-L1 vertebral level, just above the superior mesenteric artery.

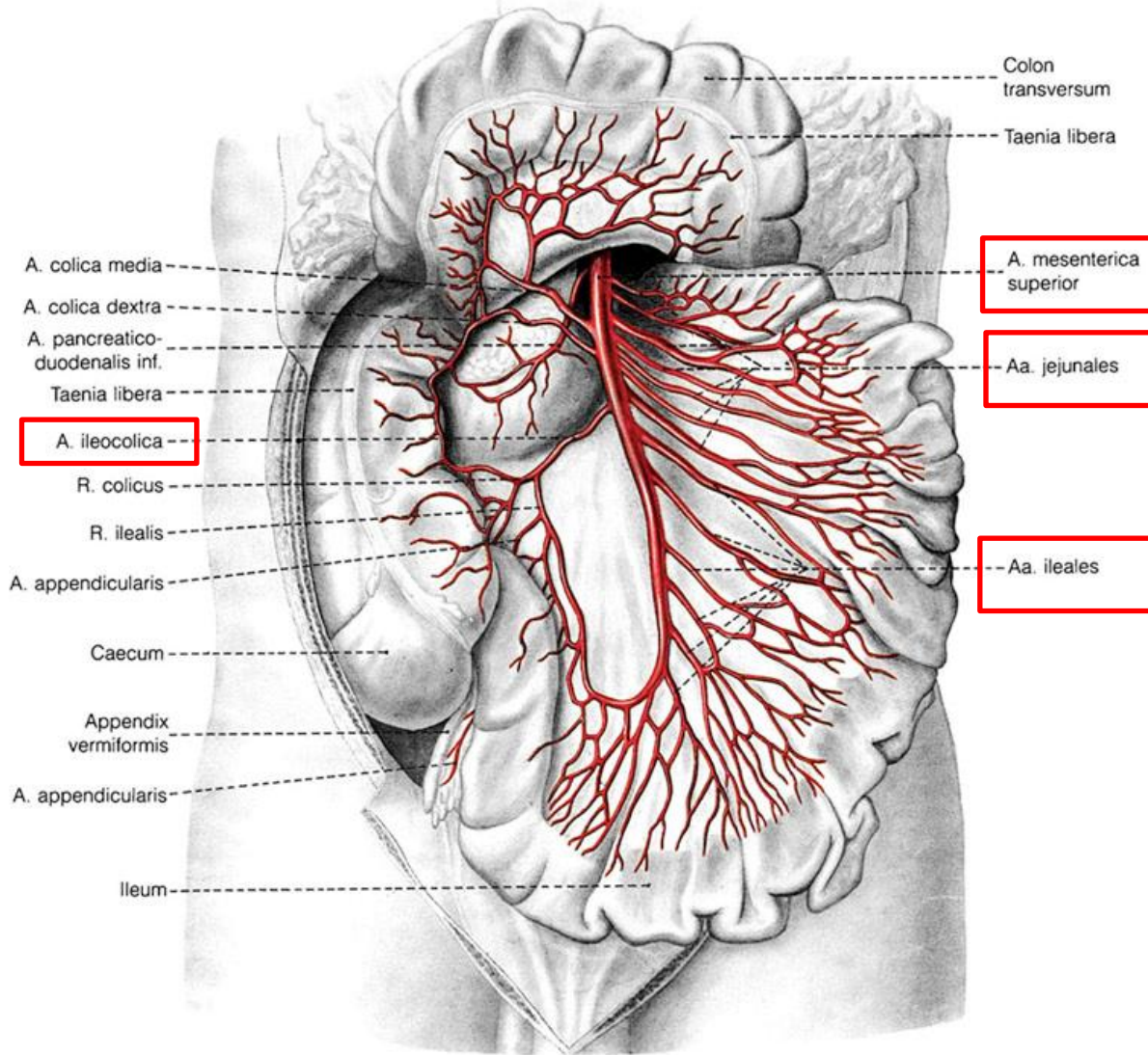


A diagram showing a cross-section of a plant stem. The stem is circular with a central pith. The vascular bundles are arranged in a ring, each consisting of a central xylem, a surrounding cambium, and an outer phloem. The diagram is labeled with 'Xylem', 'Cambium', and 'Phloem'.

antimesenterial



Jejunum and ileum – blood supply



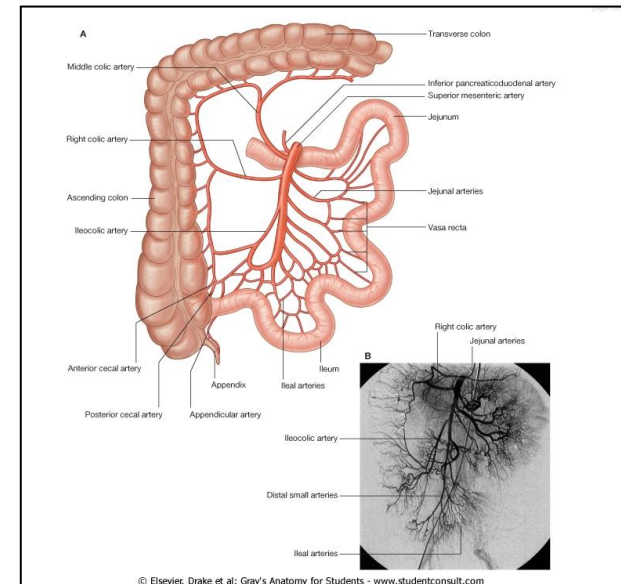
Branches of the sup. mes. a.

From the **superior mesenteric a.**

- 4-5 jejunal aa.
- approx. 12 ileal aa.
- Ileocolic a.

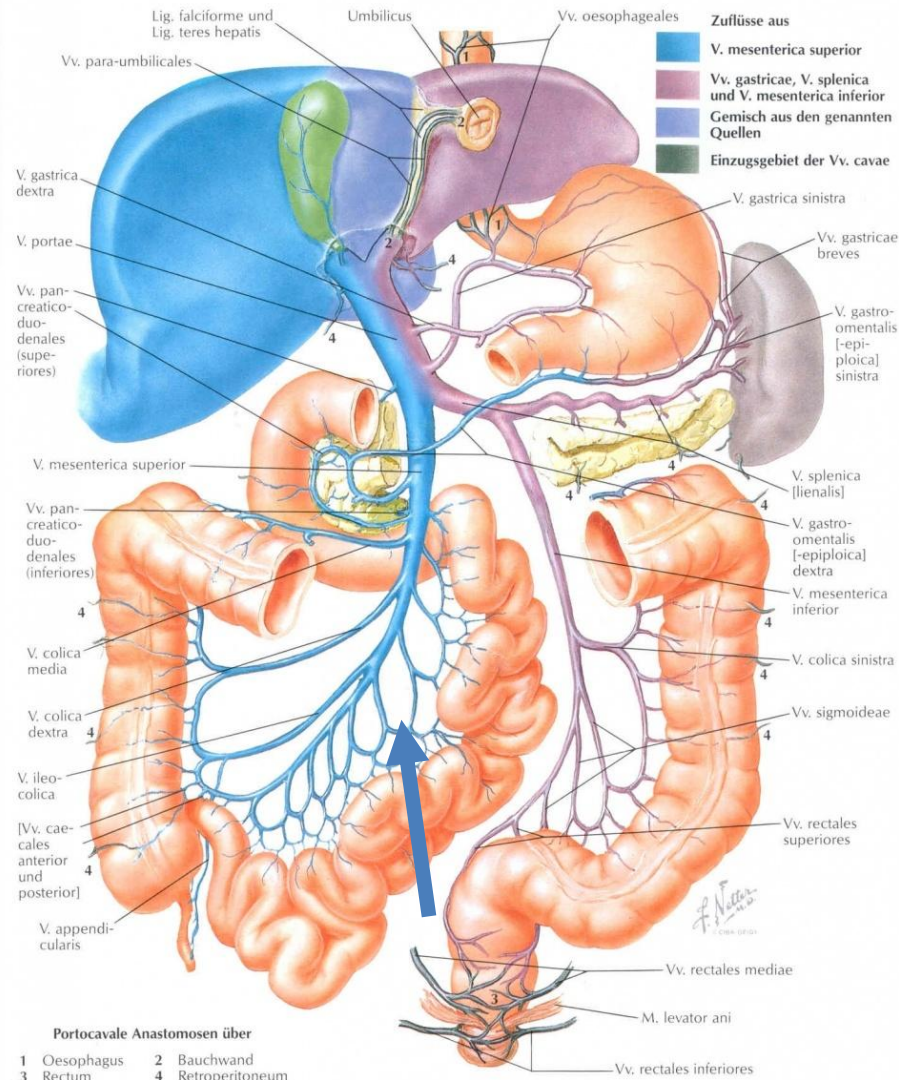
Primary
Secodary
Tertiary } arterial
 arcades

Vasa recta: the end branches

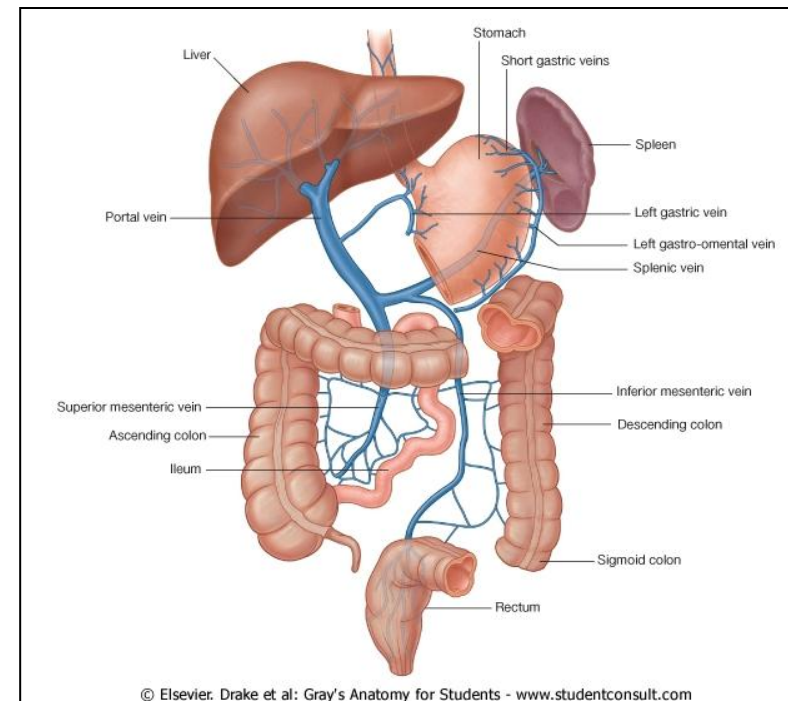
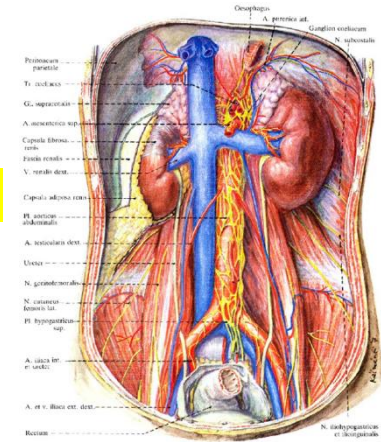


Venous drainage of the jejunum and ileum

Jejunal and ileale vv. → to sup. mesenteric v. → to hepatic portal vein

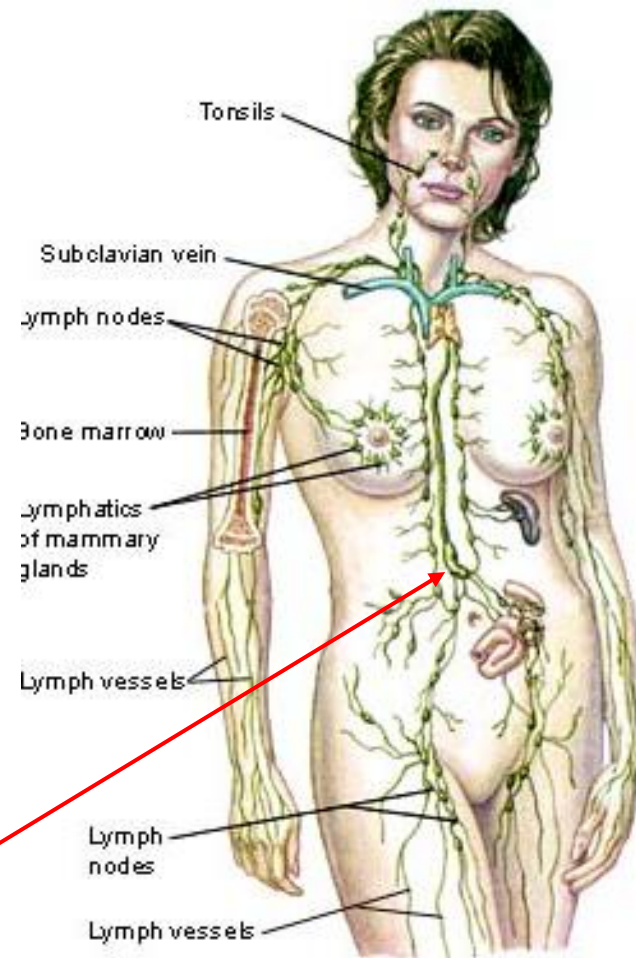
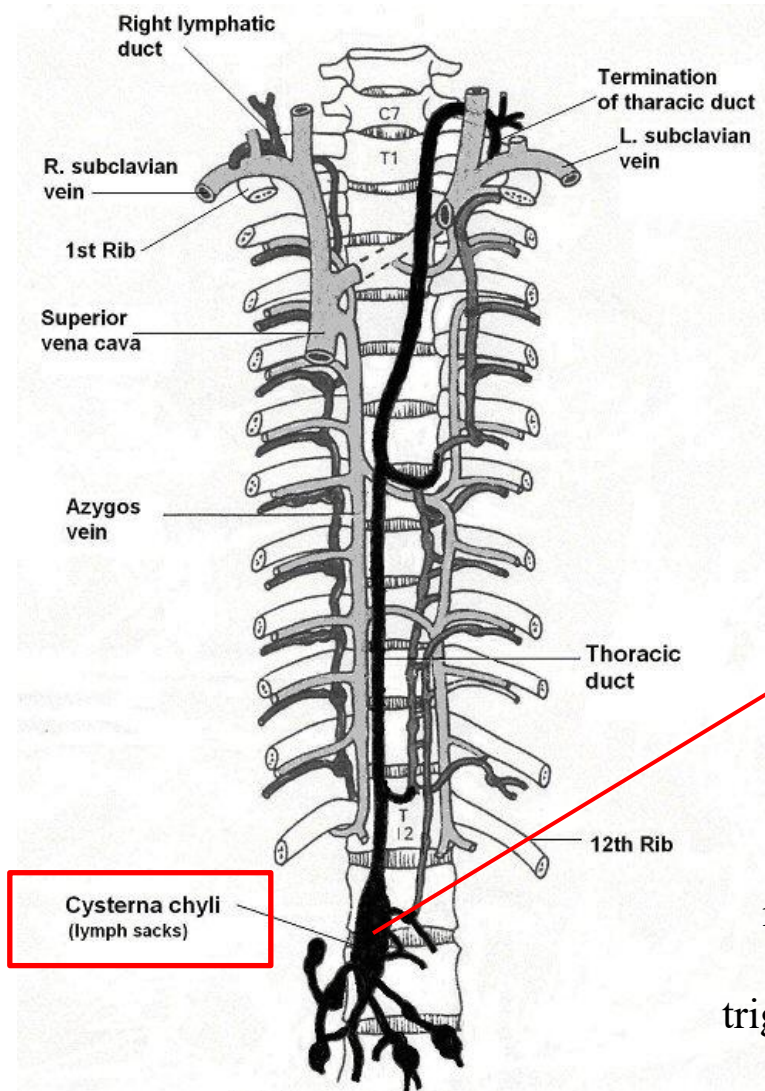


NOT (!) into the IVC



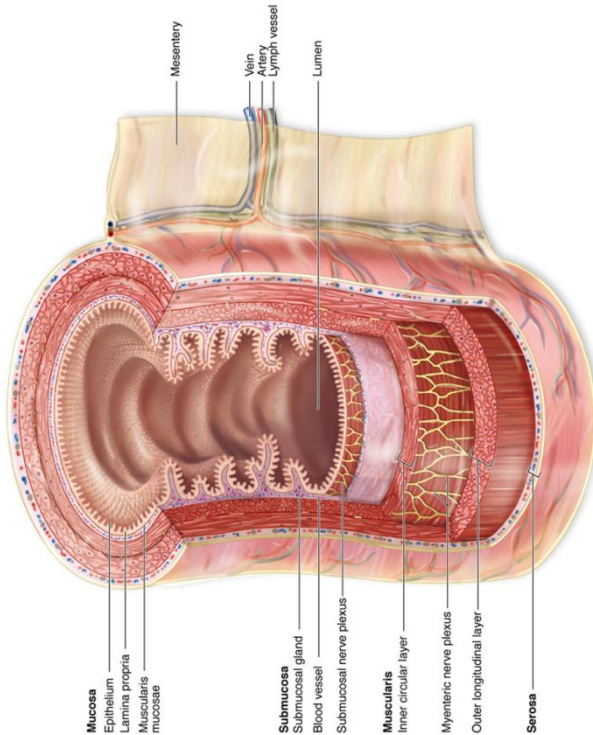
Lymphatic drainage

Cysterna chyli and thoracic duct

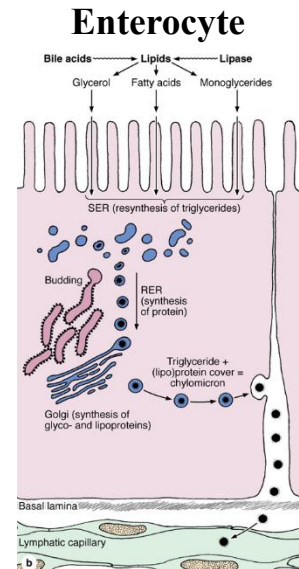
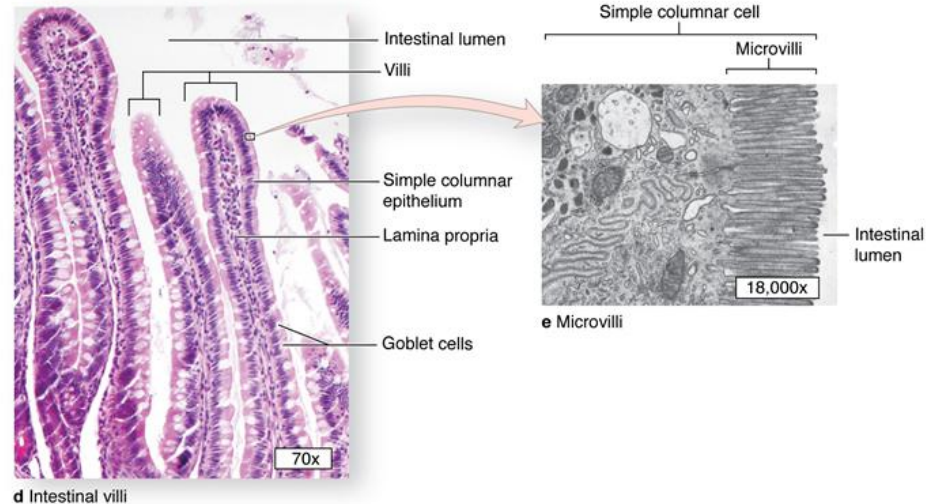
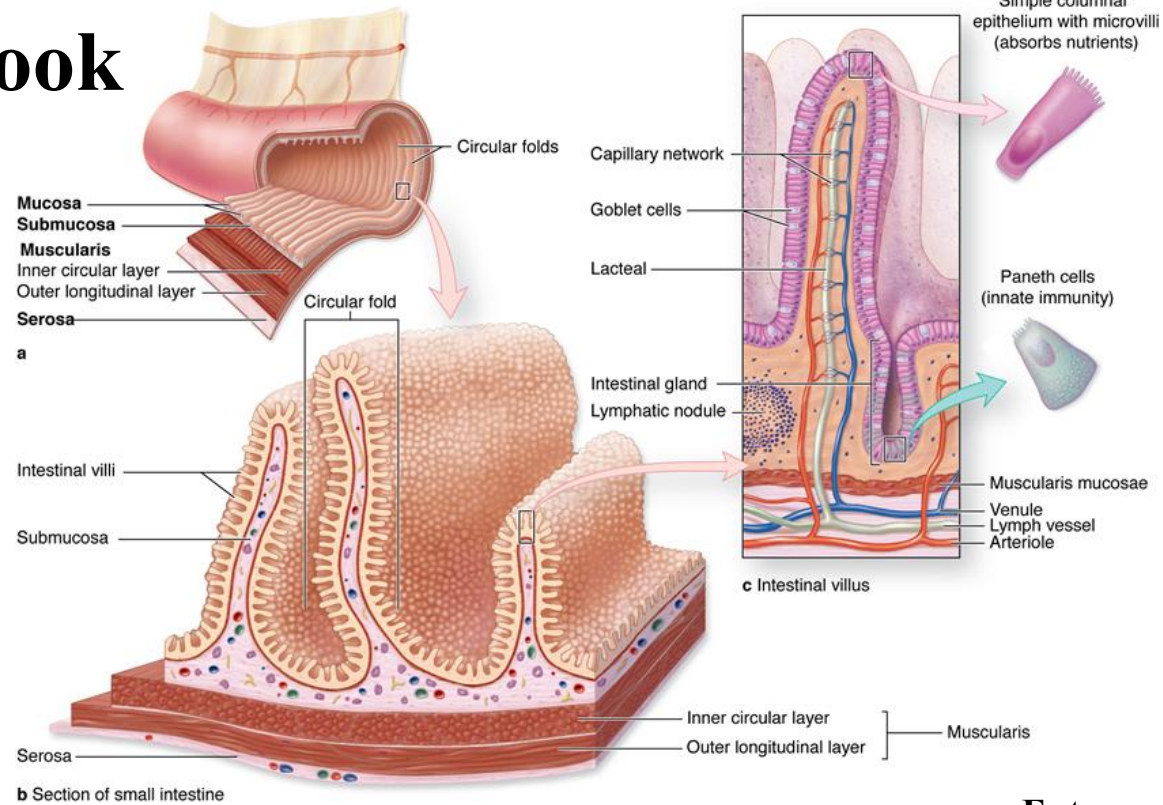


Lipid absorption in the small intestine is a multi-step process: dietary fats are emulsified by bile salts, hydrolyzed by pancreatic lipase into fatty acids and monoglycerides, and packaged into micelles. In the intestinal enterocytes, where they are absorbed, re-esterified into triglycerides, packaged into **chylomicrons**, and transported via the **lymphatic system**.

Histological outlook



- Circular folds (Kerckring's)
- Intestinal villi
- Microvilli (3000/cell)



Innervation: parasympathetic (vagus) and sympathetic praevertebral ganglia (celiac ggl. & superior mes. ggl.)

